

4.1 Answer each of the following:

- a.** Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- b.** Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- c.** Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

1.

a.

$$SSJ = \sum (y_i - \bar{y})^2 = \sum y_i^2 - N \bar{y}^2$$

$$= 7825.94 - 20 \cdot 19.21^2$$

$$= 445.458$$

$$R^2 = \frac{376.47}{445.458} = 0.8429$$

b.

$$SSR = 7825.94 \times 0.7911 = 574.29$$

$$SSE = 7825.94 - 574.29 = 151.65$$

$$F^2 = \frac{SSE}{N-2} = \frac{151.65}{20-2} = 8.425$$

c.

$$SSR = 631.63 - 182.85 = 448.78$$

$$R^2 = \frac{448.78}{631.63} = 0.7105$$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- a.** the predicted value of y for $x_0 = 4$.
- b.** the $\text{se}(f)$ corresponding to part (a).
- c.** a 95% prediction interval for y given $x_0 = 4$.
- d.** a 99% prediction interval for y given $x_0 = 4$.
- e.** a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

a.

$$\hat{y}_0 = 1.2 + 0.8 \times 4 = 4.4$$

b.

$$\text{Var}(f) = \sigma^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]$$

$$= 1.2 \left(1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right)$$

$$= 2.52$$

$$\text{se}(f) = \sqrt{2.52} = 1.5875$$

$$\hat{\sigma}^2 = \frac{3.6}{5-2} = 1.2$$

$$\bar{x} = 1$$

c. 95% P.I

$$= [4.4 \pm 1.5875 \times 3.182]$$

$$= [-0.6514, 9.4514]$$

$$t_{0.975,3} = 3.182$$

d. 99% P.I

$$= [4.4 \pm 1.5875 \times 5.841]$$

$$= [-4.8726, 13.6726]$$

$$t_{0.995,3} = 5.841$$

e.

$$\text{Var}(f) = 1.2 \left(1 + \frac{1}{5} + \frac{(1-1)^2}{10} \right) = 1.44$$

$$\text{se}(f) = \sqrt{1.44} = 1.2$$

$$\hat{y} = 1.2 + 0.8 \cdot 1 = 2$$

95% P.I

$$= [2 \pm 1.2 \times 3.182]$$

$$= [-1.8184, 5.8184]$$

(c) 區間寬度 = $9.4514 - (-0.6514) = 10.1028$

(e) 區間寬度 = $5.8184 - (-1.8184) = 7.6368$

$\therefore X = \bar{x}$ 的 P.I 寬度較窄

因為預測是距離樣本平均值越近，變異性越小，所以區間越窄

4.25 The file *collegetown* contains data on 500 houses sold in Baton Rouge, LA during 2009–2013. Variable descriptions are in the file *collegetown.def*.

- a. Estimate the log-linear model $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.
- b. Estimate the log-log model $\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$. Interpret the estimated parameters. Calculate the slope and elasticity at the sample means, if necessary.
- c. Compare the R^2 value from the linear model $PRICE = \delta_1 + \delta_2 SQFT + e$ to the “generalized” R^2 measure for the models in (b) and (c).
- d. Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?
- e. For each of the models in (a)–(c), plot the least squares residuals against *SQFT*. Do you observe any patterns?
- f. For each model in (a)–(c), predict the value of a house with 2700 square feet.
- g. For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.
- h. Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

a. Table: Regression output showing the coefficients

	Estimate	Std..Error	t.value	Pr...t..
----- ----- ----- -----				
(Intercept)	4.3938658	0.0432883	101.50231	0
sqft	0.0360445	0.0014856	24.26285	0

$$\ln(\hat{\text{price}}) = 4.3939 + 0.036 \cdot \text{sqft}$$

$\beta_2 = 0.036$, 当 sqft 每增加 1 单位时, 房价平均会增加 3.6 %

$$\text{mean price} = 250.24 \quad \text{mean sqft} = 27.28$$

$$\text{slope} = \beta_2 \cdot \text{price} = 0.036 \times 250.24 = 9.0086$$

$$\text{elasticity} = \beta_2 \cdot \text{sqft} = 0.036 \times 27.28 = 0.9821$$

$$R^2 = 0.5417$$

b. Table: Regression output showing the coefficients

	Estimate	Std..Error	t.value	Pr...t..
----- ----- ----- -----				
(Intercept)	2.049653	0.1579704	12.97492	0
log(sqft)	1.024828	0.0483922	21.17754	0

$$\ln(\hat{\text{price}}) = 2.0499 + 1.0248 \ln(\text{sqft})$$

$\beta_2 = 1.0248$, 当 sqft 每增加 1 % 时, 房价平均会增加 1.0248 %.

$$\text{slope} = \frac{\beta_2 \cdot \text{price}}{\text{sqft}} = \frac{1.0248 \times 250.24}{27.28} = 9.4005$$

$$\text{elasticity} = \beta_2 = 1.0248$$

$$R^2 = 0.4738$$

c.

不同 model 的 R^2 不能直接比较, 因此要先转换成 generalized R^2

i. linear model 的 dependent variable: price, log-linear model 和 log-log model 的 dependent variable: $\ln(\text{price})$.

ii. 将 log-linear 和 log-log model 的预测值转换成 price

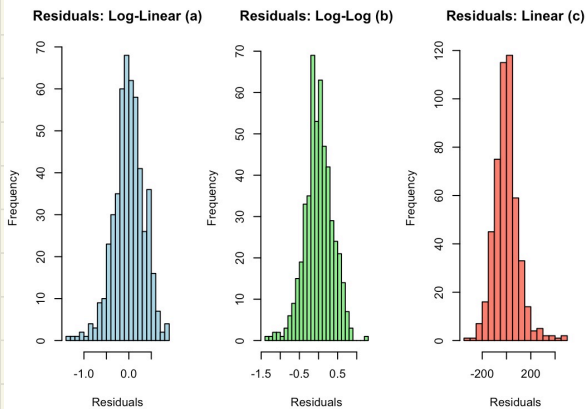
Table: Generalized R^2

gen_r2_1	gen_r2_2	r2_linear
0.6621612	0.6445084	0.6413167

由此可知, log-linear model 的 generalized $R^2 = 0.6622$ 最大.

表示这三个 model 中, log-linear model 的配适度最好, 可以解释房价中 66.22% 的变差. 透过对数转换, 提高了房价能变异的解释力.

d.



Jarque Bera Test

data: resid(mod1)
X-squared = 26.679, df = 2, p-value = 1.61e-06

Jarque Bera Test

data: resid(mod2)
X-squared = 14.279, df = 2, p-value = 0.0007933

Jarque Bera Test

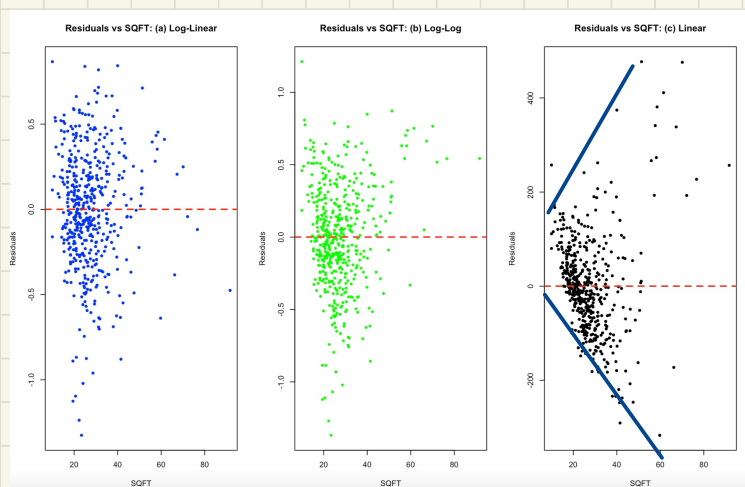
data: resid(model_c_lin)
X-squared = 221.11, df = 2, p-value < 2.2e-16

從 Jarque-Bera Test 的結果來看, 3個 model 的 $p\text{-value} < 0.05$, 所以拒絕 H_0 , 表示殘差不服從常態分配,

從直方圖來看, linear model 的 residual 有明顯的右偏,

經過對數轉換後, log-linear model & log-log model 的 residual 分佈較為對稱, 改善了右偏的問題。

e,



linear model 可以看出有明顯的漏斗狀, 表示殘差的變異隨著面積變大而增加, 因此存在異質變異性的問題。

log-linear & log-log model 中, 對 price 取對數, 殘差的變異沒有隨著面積變大而明顯增加, 表示改善了異質變異性的問題。

f.

	Model	Raw exp(y_hat)	Corrected Forecast
1	Linear	246.46	246.46
2	Log-Linear	214.23	226.98
3	Log-Log	227.54	243.15

g.

	Model	Lower Bound	Upper Bound
1	Linear	44.28	448.63
2	Log-Linear	109.77	418.12
3	Log-Log	111.14	465.84

h. 會選擇 log-linear model

1. 如果依照 generalized R^2 , 因為其 generalized R^2 最大

2. 依 (d) (e) 的殘差圖來看, log-linear model 改善了異變度異誤和右偏的問題.

5.5 For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0$.

- a.** $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2/T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b.** $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}$$

- c.** Provide an intuitive explanation for these results.

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- a.** $\beta_2 = 0$
b. $\beta_1 + 2\beta_2 = 5$
c. $\beta_1 - \beta_2 + \beta_3 = 4$

5.5

$$a. \text{Var}(b_2|X) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2} = \frac{\sigma^2}{\sum (t_i - \bar{t})^2}$$

$$\begin{aligned} \sum (t_i - \bar{t})^2 &= \sum t_i^2 - \frac{(\sum t_i)^2}{T} \\ &= \frac{T(T+1)(2T+1)}{6} - \frac{T^2(T+1)^2/4}{T} \\ &= \frac{2T(T+1)(2T+1) - 3T(T+1)^2}{12} \\ &= \frac{4T^3 + 6T^2 + 2T - 3T^3 - 6T^2 - 3T}{12} = \frac{T^3 - T}{12} \end{aligned}$$

$$\text{Var}(b_2|X) = \frac{\sigma^2}{(T^3 - T)/12} = \frac{12\sigma^2}{T^3 - T}$$

$$\text{As } T \rightarrow \infty, T^3 - T \rightarrow \infty, \lim_{T \rightarrow \infty} \text{Var}(b_2|X) = 0.$$

Thus the least squares estimator is consistent. #

$$b. x = (0.5)^t \quad r = 0.5$$

$$\sum x = \sum (0.5)^t = \frac{0.5(1 - 0.5^T)}{1 - 0.5} = 1 - 0.5^T$$

$$\sum x^2 = \sum (0.5^t)^2 = \sum 0.25^t = \frac{0.25(1 - 0.25^T)}{1 - 0.25} = \frac{1}{3}(1 - 0.25^T)$$

$$\sum (t - \bar{t})^2 = \frac{1}{3}(1 - 0.25^T) - \frac{(1 - 0.5^T)^2}{T}$$

$$\text{Var}(b_2|x) = \frac{\sigma^2}{\frac{1}{3}(1-0.25^T) - \frac{(1-0.5^T)^2}{T}}$$

$$\text{As } T \rightarrow \infty, 0.25^T \rightarrow 0, 0.5^T \rightarrow 0$$

$$\therefore \lim_{T \rightarrow \infty} \frac{1}{3}(1-0.25^T) - \frac{(1-0.5^T)^2}{T} = \frac{1}{3}$$

$$\Rightarrow \lim_{T \rightarrow \infty} \text{Var}(b_2|x) = \frac{\sigma^2}{\frac{1}{3}} = 3\sigma^2 \neq 0$$

The least squares estimator isn't consistent.

C, For an estimator to be consistent, increasing sample size (T) gives more information about β_2 , so total variation we be infinite.

$$(a) \text{ As } T \rightarrow \infty, \text{ total variation} \rightarrow \infty, \text{Var}(b_2|x) = 0$$

So it is consistent.

$$(b) \text{ As } T \rightarrow \infty, x_t = 0.5^t \rightarrow 0, \text{ total variation} = \frac{1}{3}.$$

Increasing in T doesn't provide any information about β_2 .

So it isn't consistent.

5.6

a. $se(b_2) = \sqrt{4} = 2$

1. $H_0: \beta_2 = 0$ v.s $H_1: \beta_2 \neq 0$

2. $t = \frac{b_2 - 0}{se(b_2)} \sim t(100)$ n 是多大, 近似正 normal distribution.

3. $\alpha = 0.05$ $t_{0.975}(100) = 2$

$$RR = \{ |t| > 2 \}$$

4. $t^* = \frac{3}{2} = 1.5 \notin RR$

5. Because $t^* = 1.5$ doesn't fall in reject region, we fail to reject H_0 . There isn't evidence to support H_1 is true

b. $se(b_1 + 2b_2) = \sqrt{3 + 2^2 \cdot 4 + 4 \cdot (-2)} = \sqrt{11} = 3.3166$

1. $H_0: \beta_1 + 2\beta_2 = 5$ v.s $H_1: \beta_1 + 2\beta_2 \neq 5$

2. $t = \frac{(b_1 + 2b_2) - 5}{se(b_1 + 2b_2)} \sim t(100)$

3. $\alpha = 0.05$ $t_{0.975}(100) = 2$

$$RR = \{ |t| > 2 \}$$

4. $t^* = \frac{(2+2.3) - 5}{3.3166} = 0.9045 \notin RR$

5. Because $t^* = 0.9045$ doesn't fall in reject region, we fail to reject H_0 . There isn't evidence to support H_1 is true.

6. $se(b_1 - b_2 + b_3) = \sqrt{3+4+3-2 \cdot 1 \cdot 2 + 2 \cdot 1 \cdot 2 \cdot 0} = 4$

1. $H_0: \beta_1 - \beta_2 + \beta_3 = 4$ v.s. $H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$

2. $t = \frac{(b_1 - b_2 + b_3) - 4}{se(b_1 - b_2 + b_3)} \sim t(60)$

3. $\alpha = 0.05$ $t_{0.995}(60) = 2$

$RR = \{ |t| > 2 \}$

4. $t^* = \frac{(2-3-1) - 4}{4} = -1.5 \notin RR$

5. Because $t^* = -1.5$ doesn't fall in reject region, we fail to reject H_0 . There isn't evidence to support H_1 is true

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \dots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x . Given that assumptions MR1–MR6 hold:

- a. Show that $\hat{\beta}_{2,mean}$ is unbiased.
- b. Derive an expression for $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$.
- c. Write down an expression for $\text{var}(\hat{\beta}_{2,mean})$.
- d. Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2 ?
- e. Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:
 - i. $\text{var}(b_2 | \mathbf{x})$ and $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$ where b_2 is the OLS estimator.
 - ii. Estimates for $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ where s_x^2 is the sample standard deviation for x using N as a divisor.
- f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?
- g. Show that $E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{p} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)^2/12$.]
- h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

5.20

$$a. \quad \bar{y}_2 - \bar{y}_1 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2 - (\beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1) = \beta_2 (\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)$$

$$\hat{\beta}_{2, \text{mean}} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$$E(\hat{\beta}_{2, \text{mean}} | X) = \beta_2 + E\left(\frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} | X\right) = \beta_2 + \frac{E(\bar{e}_2 | X) - E(\bar{e}_1 | X)}{\bar{x}_2 - \bar{x}_1} = \beta_2$$

$$\because E(e_i | X) = 0 \quad \therefore E(\bar{e}_1 | X) = E(\bar{e}_2 | X) = 0$$

$\Rightarrow \hat{\beta}_{2, \text{mean}}$ is unbiased. #

$$b. \quad \text{Var}(\bar{e}_2 - \bar{e}_1 | X) = \text{Var}(\bar{e}_2 | X) + \text{Var}(\bar{e}_1 | X) = \frac{2}{N} \sigma^2 + \frac{2}{N} \sigma^2 = \frac{4}{N} \sigma^2$$

$$\text{Var}(\bar{e}_1 | X) = \text{Var}\left(\frac{1}{N/2} \sum e_i\right) = \frac{4}{N^2} \cdot \frac{N}{2} \sigma^2 = \frac{2}{N} \sigma^2$$

$$\text{同理 } \text{Var}(\bar{e}_2 | X) = \frac{2}{N} \sigma^2$$

$$\text{Var}(\hat{\beta}_{2, \text{mean}} | X) = \text{Var}\left(\frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} | X\right) = \left(\frac{1}{\bar{x}_2 - \bar{x}_1}\right)^2 \text{Var}(\bar{e}_2 - \bar{e}_1 | X)$$

$$= \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \quad \#$$

$$\text{Var}(b_2 | X) = \frac{\sigma^2}{N \sum (x_i - \bar{x})^2}$$

$$c, \quad \text{Var}(Y) = E[\text{Var}(Y|X)] + \text{Var}[E(Y|X)]$$

$$\text{Var}(\hat{\beta}_{2,\text{mean}}) = E[\text{Var}(\hat{\beta}_{2,\text{mean}}|X)] + \text{Var}[E(\hat{\beta}_{2,\text{mean}}|X)]$$

$$= E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] + \text{Var}(\beta_2)$$

"0" $\because \beta_2$ is constant

$$= E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right] \quad \#$$

d. consistency: $\lim_{N \rightarrow \infty} \text{Var}(\hat{\beta}_{2,\text{mean}}|X) = 0$

只要 X 存在變異, 由 $N \rightarrow \infty$, $N(\bar{x}_2 - \bar{x}_1)^2 \rightarrow \infty$

$$\therefore \lim_{N \rightarrow \infty} \text{Var}(\hat{\beta}_{2,\text{mean}}|X) = 0$$

$\hat{\beta}_{2,\text{mean}}$ is consistent estimator for β_2 .

e,

	N	var_b2	var_b2_mean	est_E_s2_inv	est_E_diff_inv
1	100	1.2436	1.6632	0.1244	0.1663
2	500	0.2440	0.3260	0.1220	0.1630
3	1000	0.1229	0.1656	0.1229	0.1656
4	5000	0.0241	0.0323	0.1207	0.1616

f. 隨著 N 增加, $\text{Var}(b_2|X)$ 和 $\text{Var}(\hat{\beta}_{2,\text{mean}}|X)$ 皆下降, 只要 N 夠

大就會趨近於 0. $\text{Var}(b_2|X)$ 和 $\text{Var}(\hat{\beta}_{2,\text{mean}}|X)$ 皆具備一致性,

在所有 N 下, $\text{Var}(b_2|X) < \text{Var}(\hat{\beta}_{2,\text{mean}}|X)$, 符合 Gauss-Markov

Theorem, 在滿足古典線性假設下, OLS 是 BLUE.

g.

$$X \sim U(0, 10)$$

$$\text{Var}(X) = \frac{(b-a)^2}{12} = \frac{10^2}{12} = 8.3333$$

$$E[(S_x^2)^{-1}] = \frac{1}{8.3333} = 0.12$$

$$\bar{x}_1 \rightarrow 2.5 \quad \bar{x}_2 \rightarrow 7.5$$

$$E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \rightarrow \frac{4}{(7.5 - 2.5)^2} = 0.16 \quad \#$$

h. 若是沒有進行排序，而是隨機配對， $E(\bar{x}_2 - \bar{x}_1) = 0$ ，分母會接近 0

$$\text{Var}(\hat{\beta}_2, \text{mean}(X)) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \rightarrow 0$$

所以當 $N \rightarrow \infty$ ，變異數不會收斂到 0。因此不具備一致性。